$$\phi_n(x): \left\{ \frac{1}{\sqrt{2L}},\ \frac{1}{\sqrt{L}}\cos\frac{n\pi x}{L},\ \frac{1}{\sqrt{L}}\sin\frac{n\pi x}{L} \right\}$$

orthonormal basis

$$\int_{-L}^{L} \phi_n^*(x)\,\phi_m(x)\,dx = \delta_{nm}$$

$$f(x) = \sum_n c_n \phi_n(x)$$

$$c_n = \int dx'\ \phi_n^*(x')\,f(x')$$

$$\sum_n \phi_n^*(x)\,\phi_n(x') = \delta(x-x')$$

completeness relation

define $\quad G(x,x') = \sum_n \phi_n^*(x)\,\phi_n(x')$

$$f(x) = \sum_n c_n \phi_n(x)$$

$$= \sum_n \int dx'\ \phi_n^*(x')\,f(x')\,\phi_n(x)$$

$$= \int dx' \underbrace{\left(\sum_n \phi_n^*(x')\,\phi_n(x) \right)}_{G(x,x')} f(x')$$

pretty smart pick

$$f(x) = \delta(x-z)$$

$$\delta(x-z) = \int dx' \, G(x,x')\delta(x'-z)$$

$$= G(x,z) \underbrace{\int dx' \delta(x'-z)}_{=1}$$

$$\delta(x-z) = G(x,z)$$

$$\delta(x-x') = G(x,x')$$

一个函数 $f(x)$ 展开，用展开系数都带回来还原 $f(x)$

只有完备基才能做到

如果不完备，将无法还原 $f(x)$（信息变丢失）

$f(x) \to f(x)$，要有 completeness basis 作为假设